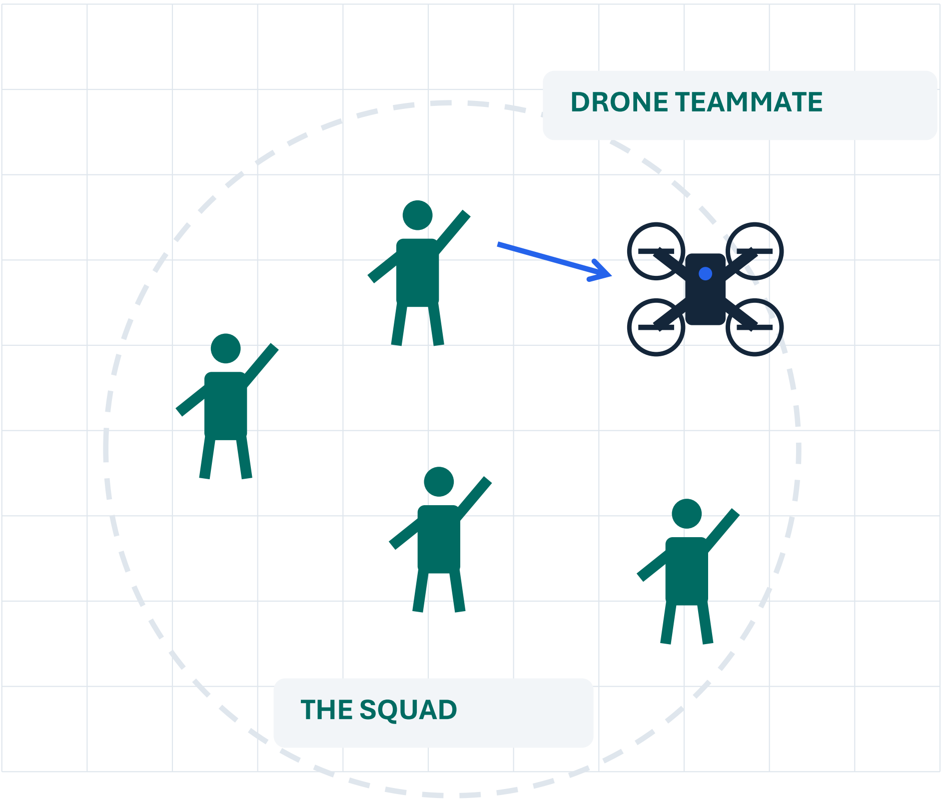
DEFENSE / WORKING SIMULATION

Add the drone to the squad.

Plan the mission. Signal changes as the squad moves.

ONE SQUAD. SHARED DRONE SUPPORT.

WHY NOW: Army drone specialists may join squads. [1]



ILLUSTRATIVE SQUAD CONCEPT

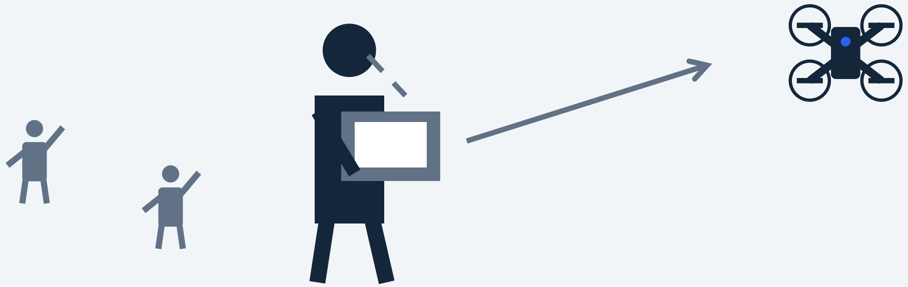
[1] Task & Purpose · Aug. 31, 2026

The squad needs to adapt. Eyes stay up.

OPERATING CONTEXT: GPS-DENIED ENVIRONMENTS

WORKFLOW DESCRIBED TO US

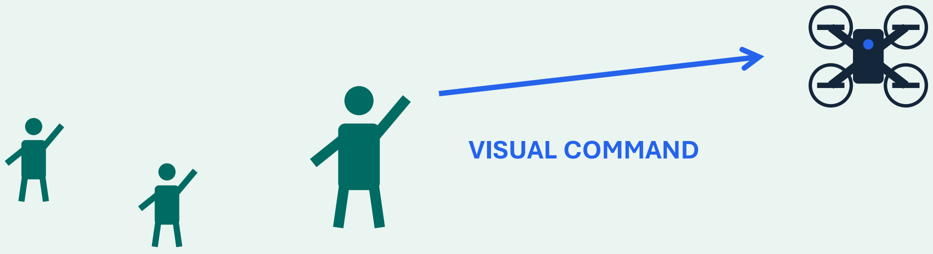
Small, chest-mounted tablet



Attention moves to the interface.

TACTLINK / ONBOARD VISION CONCEPT

Signal directly to the drone



Change intent while facing the squad.

INTENDED IMPACT

Redirect drone support as squad needs change.

Discovery: team conversation with a Mach Industries representative (paraphrased; informal user input).

Today: connected simulation. Onboard vision is planned; GPS-independent navigation is a separate requirement.

Human intent, in more than one form.

THE HACKATHON PROMPT

“Flight paths in plain English”

Plan and adjust a flight path using natural language.

→

OUR EXTENSION

Why not communicate with hand gestures, too?

Another human way to express intent and change drone behavior.

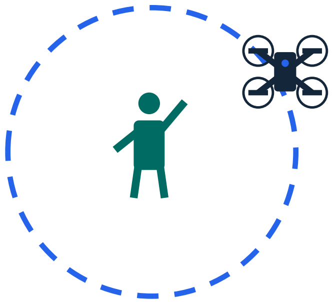
Natural-language planning inspired us. Gesture interaction became our working demonstration.

Source: DNHacks Defense inspiration prompt, “Flight paths in plain English” (provided by team).

Four commands. One continuous demo.

01

Orbit



Circle the operator.

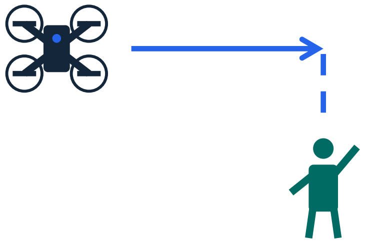
02

Move

Follow directional intent.

03

Return



Overhead to the sender.

04

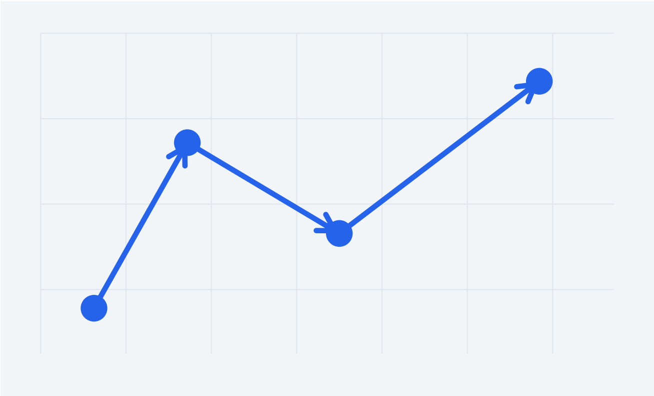
Stop

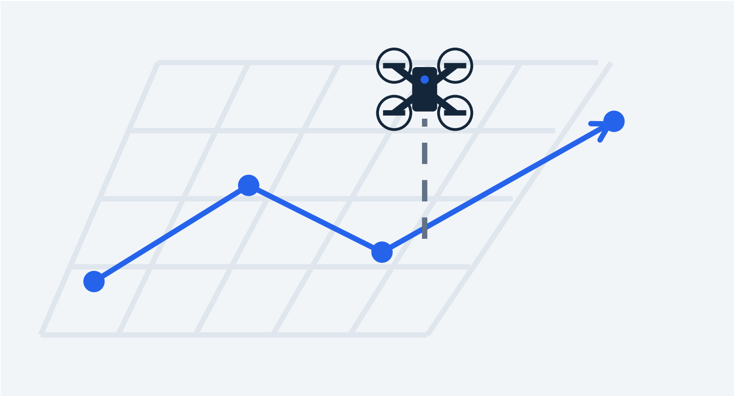
Open-palm emergency stop.

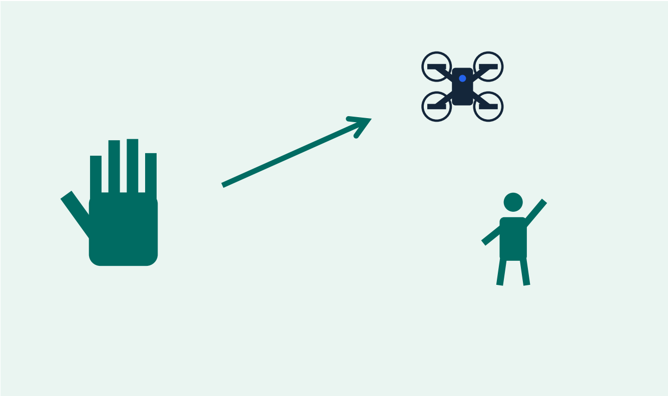
LIVE SETUP Phones as operators → Main computer → Shared 3D simulation

Plan the route. Adapt mid-flight.

A planned mission stays responsive to the operator.







01 PLAN IN 2D

Build the mission in a flight planner.

02 VISUALIZE IN 3D

See the mission in a virtual world.

03 INTERRUPT BY GESTURE

Change drone behavior mid-flight.

POSSIBLE USE CASE

Send a drone between operators to provide overwatch.

A working prototype. A testable path.

WORKS IN SIMULATION

- **Stop**
  Open-palm intervention
- **Continue**
  Assigned path or idle without a hand
- **Select**
  Preference for the nearest operator

NEXT VALIDATION GATES

01 **Measure**
Errors, latency, unintended commands

02 **Stress-test**
Visibility and competing operators

03 **Integrate**
Onboard compute and physical flight

TODAY: Connected phone prototype. No physical flight data. GPS-independent navigation remains separate.

A gesture vocabulary we can train.

We trained a model on our own hand signals.

Our custom gesture-training pipeline shows that the command vocabulary can expand beyond preset gestures.

Record examples → **Label the signals** → **Train the model** → **Connect commands**

Build the vocabulary around the people using it.

One drone.
Two ways to communicate.

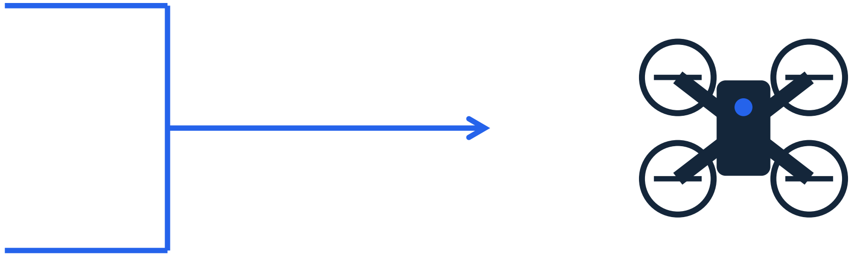
Follow the mission. Adapt through gestures and spoken commands.

HAND GESTURES

Demonstrated in simulation

“Take a picture of my location.”

SPOKEN COMMANDS / IN DEVELOPMENT



UNDERSTAND INTENT

Translate it into action